Homework 11

Solutions will be available after Monday, November 9, 2026.

Question 1: Implementing K-means

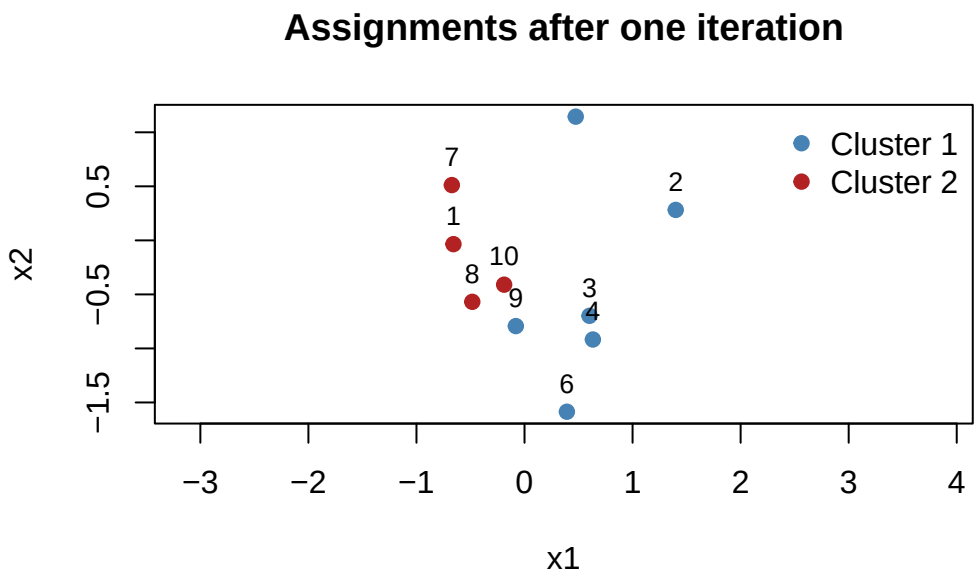
K-means is a relatively simple algorithm that we can write ourselves. For Questions 1 and 2, do not use an existing function that performs K-means directly.

Generate 10 observations with two independent standard normal coordinates. Set a random seed and report it. Randomly assign the observations to $K = 2$ clusters, making sure neither cluster is empty. Let $C(i)$ denote the cluster assigned to observation x_i, and let m_k denote the mean of cluster k.

a. Perform one iteration: calculate the cluster means, then assign each observation to its closest mean using Euclidean distance. Report the means and updated assignments, and plot the observations colored by their updated assignments.

Cluster	x1	x2
1	0.3946	-0.3718
2	-0.4481	-0.1557

Observation	Initial	Updated
1	1	2
2	1	1
3	1	1
4	1	1
5	1	1
6	1	1
7	2	2
8	2	2
9	1	1
10	2	2



I used seed 432 to generate the 10 observations and their initial assignments. The initial cluster means were $(0.3946, -0.3718)$ for cluster 1 and $(-0.4481, -0.1557)$ for cluster 2. After assigning each observation to the closest of these means, cluster 1 contained observations 1, 7, 8, and 10, while cluster 2 contained observations 2, 3, 4, 5, 6, and 9. The plot shows these updated assignments.

b. Write a function that repeats these two steps until the assignments stop changing or a specified maximum number of iterations is reached. Your function should work for a data matrix with any number of variables and a chosen number of clusters K. Start from a nonempty assignment to each cluster; if a cluster becomes empty during iteration, keep its previous mean. Recalculate the cluster means from the final assignments before returning them. Return the final assignments, cluster means, and within-cluster sum of squares,

$$\sum_{k=1}^{K} \sum_{i:\, C(i)=k} \|x_i - m_k\|^2.$$

The function alternates between calculating cluster means and assigning each observation to its closest mean. It stops when the assignments no longer change or the iteration limit is reached. If a cluster becomes empty, the function keeps its previous mean. Before returning the results, it updates the means from the final assignments and sums the squared distances from each observation to its assigned mean.

c. Apply your function to the same 10 observations from two different random initial assignments. Report the final plots, cluster means, and within-cluster sums of squares. Do the two runs produce the same grouping and objective value? Explain your findings.

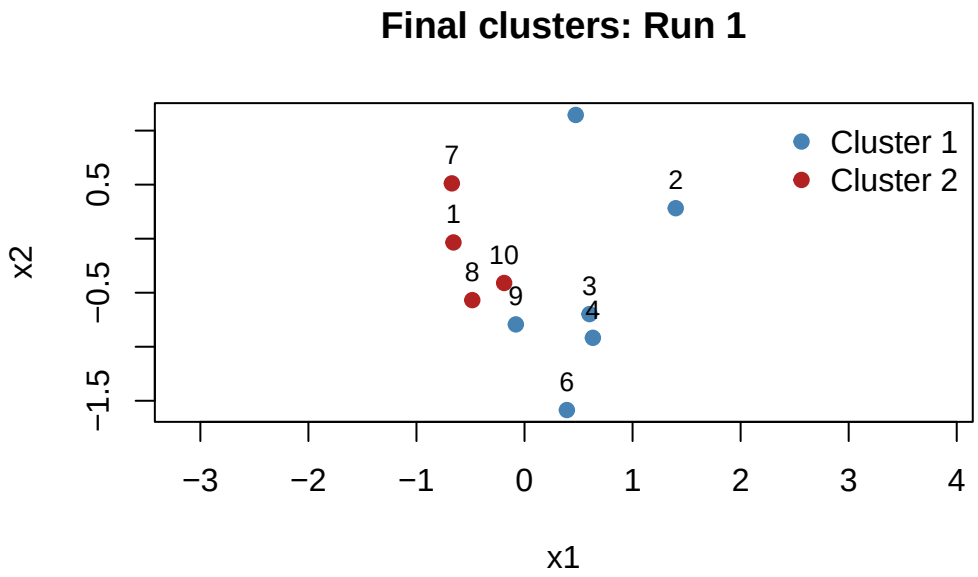
Run	Cluster	x1	x2	Within_SS
1	1	0.5700	-0.4280	5.9220
1	2	-0.5005	-0.1255	0.8435
2	1	0.1457	-0.8289	1.9066
2	2	0.1359	0.4760	3.7417

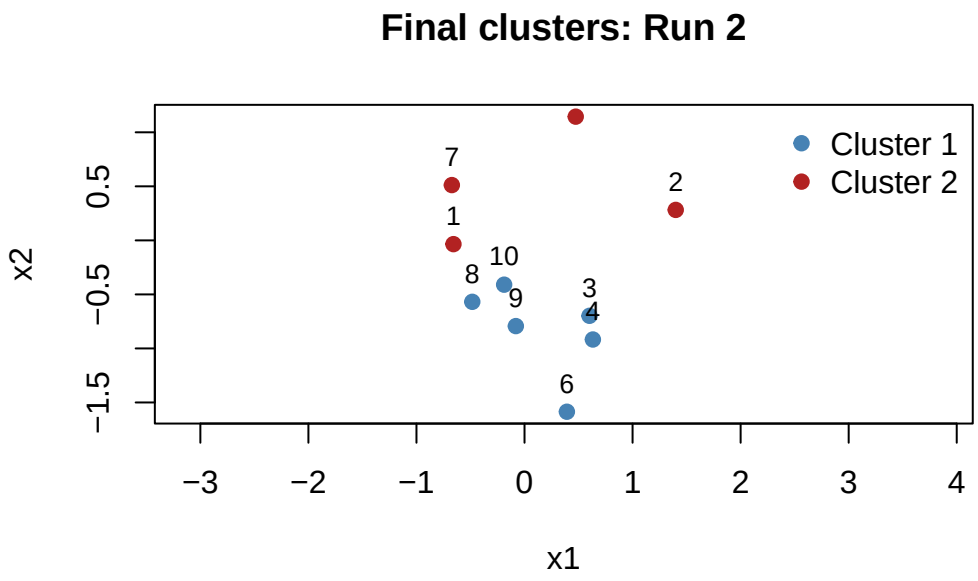
Run	Total_within_SS
1	6.7655
2	5.6483

Run	Total_within_SS

Observation	Run_1	Run_2
1	2	2
2	1	2
3	1	1
4	1	1
5	1	2
6	1	1
7	2	2
8	2	1
9	1	1
10	2	1

```
[1] FALSE
```





I reused the initial assignments from part (a) for run 1 and generated a different set with seed 433 for run 2. Both runs used the same 10 observations. They converged to different groups, even after allowing for the cluster labels to be swapped. For example, observations 1 and 2 ended up in different clusters in run 1 but in the same cluster in run 2.

The total within-cluster sum of squares was 6.7655 for run 1 and 5.6483 for run 2. Run 2 gave the tighter grouping according to the K-means objective. This shows why the starting assignments matter: the algorithm can converge to different solutions, and the better of two runs is not necessarily the overall best solution.

Question 2: Image color compression

Pick your favorite color image and use your K-means function from Question 1 to cluster its pixels. For computational efficiency, resize the image to no more than 200×200 pixels and use its three RGB color channels.

a. Plot the original image and report its dimensions. Transform it into a data matrix with one observation per pixel and three variables: red, green, and blue. Report the dimensions of this matrix.

```
Warning: package 'magick' was built under R version 4.3.3
```

```
Linking to ImageMagick 6.9.12.93
Enabled features: cairo, fontconfig, freetype, heic, lcms, pango, raw, rsvg, webp
Disabled features: fftw, ghostscript, x11
```

Image	Width	Height
Original	800	800
Resized for clustering	200	200

```
[1] 40000     3
```

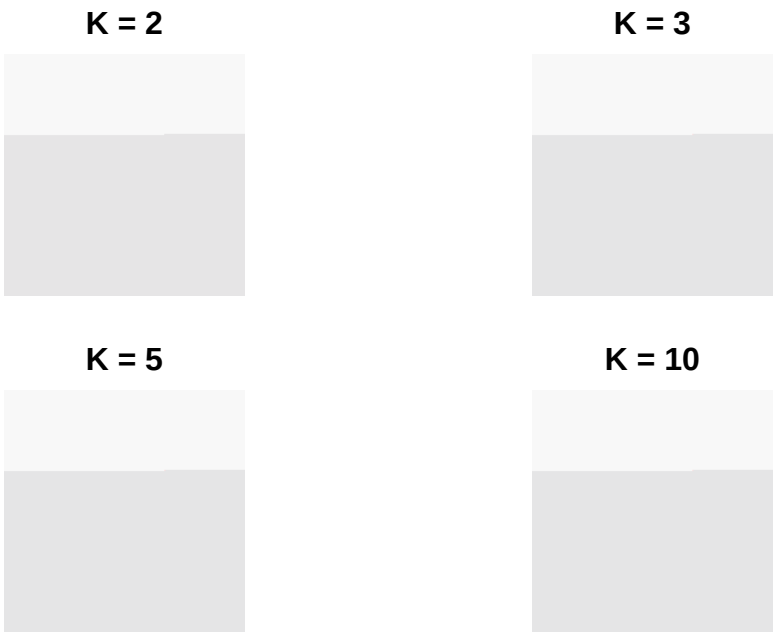
My original image was 800 × 800 pixels. I resized it to 200 × 200 pixels for clustering, giving me a data matrix with 40,000 rows and 3 columns. Each row represents one pixel, and the three columns are its red, green, and blue values on a scale from 0 to 1.

b. For $K = 2, 3, 5, 10$, apply your function to the pixel data. Report the cluster means and within-cluster sum of squares for each K. Reconstruct and plot each compressed image by replacing every pixel with its assigned cluster mean. Use the same RGB scale for all fits.

K	Total_WSS
2	7.695
3	7.689
5	7.689
10	7.689

K	Cluster	Red	Green	Blue	Within_SS
2	1	0.9725	0.9725	0.9725	0.0002
2	2	0.9000	0.9000	0.9000	7.6948
3	1	0.9000	0.9000	0.9000	7.6892
3	2	0.9725	0.8863	0.8863	0.0000
3	3	0.9725	0.9725	0.9725	0.0002
5	1	0.9241	0.9241	0.9241	0.0000
5	2	0.9000	0.9000	0.9000	7.6892
5	3	0.9725	0.8863	0.8863	0.0000
5	4	0.9244	0.9244	0.9244	0.0000
5	5	0.9725	0.9725	0.9725	0.0002
10	1	0.9725	0.8863	0.8863	0.0000
10	2	0.9244	0.9243	0.9243	0.0000
10	3	0.9241	0.9241	0.9241	0.0000
10	4	0.9240	0.9240	0.9240	0.0000
10	5	0.9241	0.9241	0.9241	0.0000
10	6	0.9246	0.9246	0.9246	0.0000
10	7	0.9243	0.9243	0.9243	0.0000

K	Cluster	Red	Green	Blue	Within_SS
10	8	0.9000	0.9000	0.9000	7.6892
10	9	0.9725	0.9725	0.9725	0.0002
10	10	0.9245	0.9245	0.9245	0.0000



I used seed 434 and ran my K-means function five times for each value of K, keeping the run with the lowest total within-cluster sum of squares (WSS). To make each compressed image, I replaced every pixel with its cluster's mean RGB color. I used the same RGB scale for all four images.

K	Total WSS
2	7.6951
3	7.6894
5	7.6894
10	7.6894

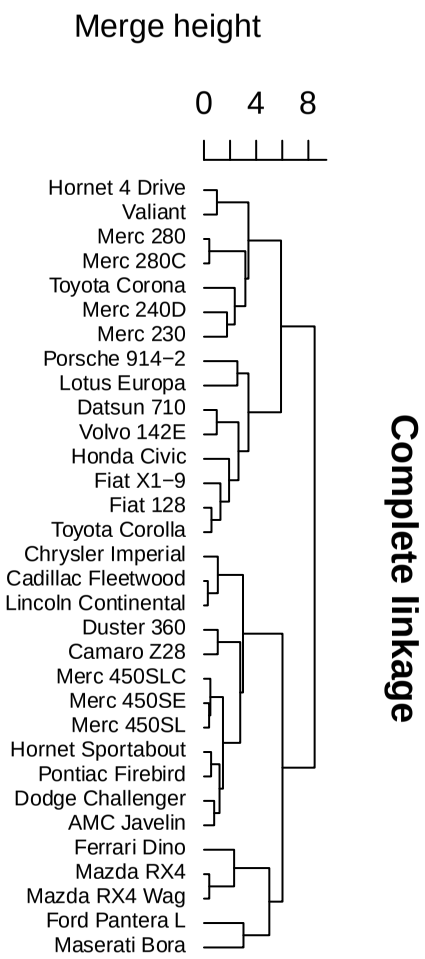
c. Which value of K would you choose for this image? Briefly justify your choice using the reconstructed images and the within-cluster sums of squares.

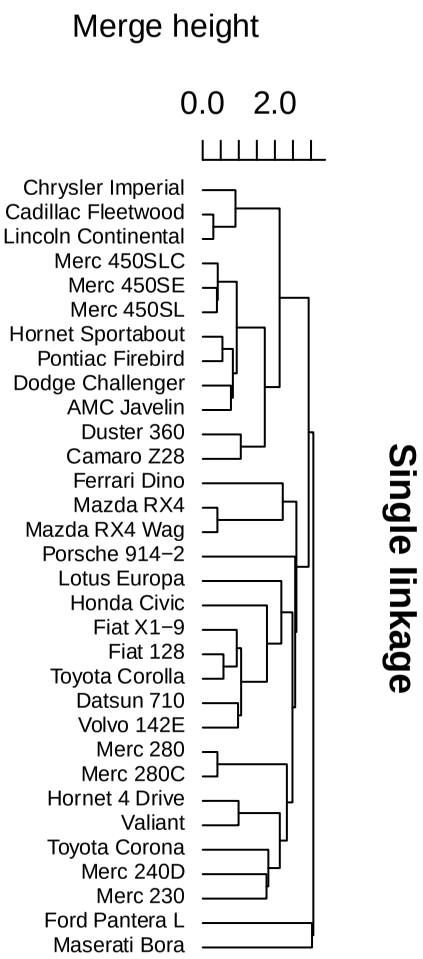
I would choose $K = 2$ because the image is already almost entirely pale pink, and the two-color version keeps its overall look. Going from $K = 2$ to $K = 3$ reduced the WSS from 7.6951 to 7.6894, a drop of less than 0.1%. The WSS stayed the same to four decimal places at $K = 5$ and $K = 10$. For this photo, I don't think the extra colors are worth using.

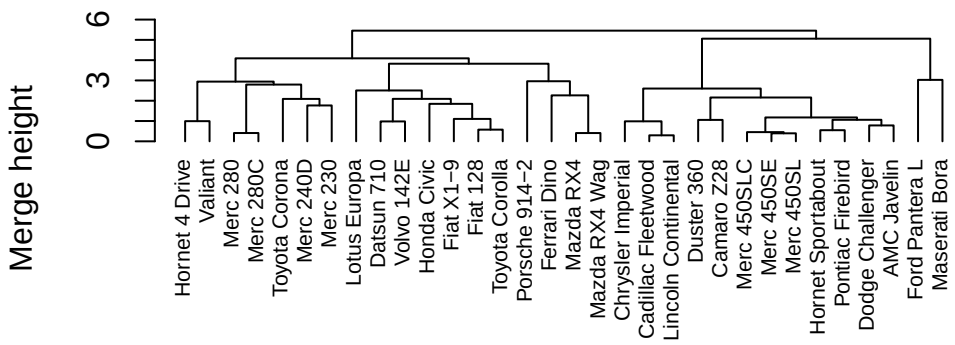
Question 3: Hierarchical clustering of cars

The file `data/mtcars.csv` contains 11 numerical variables for 32 cars, including fuel economy, horsepower, and weight. The `car` column gives the model names; use it to label observations, not as a clustering variable. You may use existing hierarchical clustering functions for this question.

a. Standardize each numerical variable to have mean 0 and standard deviation 1. Using Euclidean distance, perform hierarchical clustering with single, complete, and average linkage. Plot the three dendrograms with the car names and compare the results.


Merge height
0
4
8
Hornet 4 Drive
Valiant
Merc 280
Merc 280C
Toyota Corona
Merc 240D
Merc 230
Porsche 914−2
Lotus Europa
Datsun 710
Volvo 142E
Honda Civic
Fiat X1−9
Fiat 128
Toyota Corolla
Chrysler Imperial
Cadillac Fleetwood
Lincoln Continental
Duster 360
Camaro Z28
Merc 450SLC
Merc 450SE
Merc 450SL
Hornet Sportabout
Pontiac Firebird
Dodge Challenger
AMC Javelin
Ferrari Dino
Mazda RX4
Mazda RX4 Wag
Ford Pantera L
Maserati Bora
Complete linkage

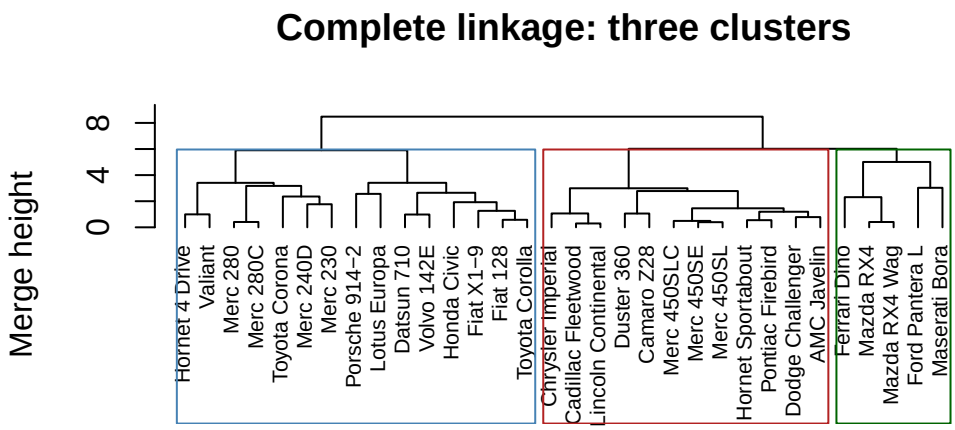

Merge height
0.0
2.0
Chrysler Imperial
Cadillac Fleetwood
Lincoln Continental
Merc 450SLC
Merc 450SE
Merc 450SL
Hornet Sportabout
Pontiac Firebird
Dodge Challenger
AMC Javelin
Duster 360
Camaro Z28
Ferrari Dino
Mazda RX4
Mazda RX4 Wag
Porsche 914−2
Lotus Europa
Honda Civic
Fiat X1−9
Fiat 128
Toyota Corolla
Datsun 710
Volvo 142E
Merc 280
Merc 280C
Hornet 4 Drive
Valiant
Toyota Corona
Merc 240D
Merc 230
Ford Pantera L
Maserati Bora
Single linkage



Method	Cluster_sizes
Single	30, 1, 1
Complete	15, 12, 5
Average	18, 12, 2

I standardized all 11 numerical variables and used Euclidean distance to compare the cars. I used the car names only as labels. At a cut of three clusters, single linkage produced one group of 30 cars and two groups with one car each. This shows the chaining effect: cars can join a large group through a series of close pairs even if some cars in the group are far apart. Complete linkage gave more balanced groups of 15, 12, and 5 cars. Average linkage gave groups of 18, 12, and 2 cars, so it kept a small group separate without showing as much chaining as single linkage.

b. Choose one linkage method and a number of clusters that you consider reasonable. Report the cluster memberships and a table of cluster means on the original variable scales. Describe the main characteristics of the groups and explain your choice using the dendrogram and table. Briefly explain why standardizing the variables matters for this analysis.



Cluster	Cars
1	Mazda RX4, Mazda RX4 Wag, Ford Pantera L, Ferrari Dino, Maserati Bora
2	Datsun 710, Hornet 4 Drive, Valiant, Merc 240D, Merc 230, Merc 280, Merc 280C, Fiat 128, Honda Civic, Toyota Corolla, Toyota Corona, Fiat X1-9, Porsche 914-2, Lotus Europa, Volvo 142E
3	Hornet Sportabout, Duster 360, Merc 450SE, Merc 450SL, Merc 450SLC, Cadillac Fleetwood, Lincoln Continental, Chrysler Imperial, Dodge Challenger, AMC Javelin, Camaro Z28, Pontiac Firebird

Cluster	Size	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
1	5	18.50	6.80	223.40	198.80	3.84	3.00	15.62	0.00	1.00	4.60	5.20
2	15	24.65	4.53	131.65	91.33	3.90	2.58	19.16	0.93	0.53	3.93	1.80
3	12	15.05	8.00	357.62	194.17	3.12	4.10	17.14	0.00	0.00	3.00	3.08

I chose complete linkage with three clusters. The dendrogram shows three branches at this cut, and the means show useful differences in fuel economy, weight, horsepower, and transmission. This cut also avoids the one-car groups that appeared with single linkage.

- **Cluster 1 (5 cars):** All five cars have manual transmissions. They averaged 18.50 mpg and had the highest average horsepower, at 198.80 hp.
- **Cluster 2 (15 cars):** These cars were the lightest on average, at about 2,580 pounds. They also had the best average fuel economy, at 24.65 mpg, and the lowest average horsepower, at 91.33 hp.
- **Cluster 3 (12 cars):** All 12 cars have eight cylinders and automatic transmissions. They were the heaviest group on average, at about 4,104 pounds, and had the lowest average fuel economy, at 15.05 mpg.

Standardizing matters because the variables have different units and spreads. For example, raw horsepower differences can be much larger numerically than weight differences recorded in thousands of pounds. Standardization makes Euclidean distance compare differences in standard deviation units. I used the original scales for the cluster means so the results are easier to interpret.